$$f(x) = y$$
$$f^{-1}(y) = x$$
$$f(f^{-1}(y)) = y$$

$$\text{let } f(a) * f(b) = f(a + b)$$
$$\text{i.e. } 10^a * 10^b = 10^{a+b}, f(x) = 10^x$$
$$\text{let } m = f^{-1}(f(a) * f(b)) = a + b$$
$$\text{let } a = f^{-1}(c)$$
$$\text{let } b = f^{-1}(d)$$
$$m = f^{-1}(c * d) = f^{-1}(c) + f^{-1}(d)$$

$$\text{let } \frac{f(a)}{f(b)} = f(a - b)$$
$$\text{let } m = f^{-1}\left(\frac{f(a)}{f(b)}\right) = a - b$$
$$\text{let } a = f^{-1}(c)$$
$$\text{let } b = f^{-1}(d)$$
$$m = f^{-1}\left(\frac{c}{d}\right) = f^{-1}(c) - f^{-1}(d)$$

the rules of a inverse function can be derived from its non-inverse and vice-versa